Shangze (Rudy) Dai

Ph.D. Candidate in Applied Economics
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Research Interests

Economics of Innovation and Technological Change; Economics of AI; Industrial Organization and Production Economics; Environmental and Resource Economics.

Education

Ph.D. in Applied Economics, University of Florida 2023 – 2027 (expected)
Research area: Economics of New Technology; GPA: 3.95/4.00

M.S. in Applied Mathematics, Johns Hopkins University 2023 – 2026
Field: Applied Analysis and Stochastic Processes; GPA: 3.84/4.00

M.Phil. in Economics, Wuhan University 2020 – 2023
Field: Regional Economics; GPA: 3.82/4.00

Certificate in International Studies, Hopkins-Nanjing Center 2022 – 2023
Field: International Economics; GPA: 3.91/4.00

B.S. in Economics, Tongji University 2016 – 2020
Field: Mathematical Finance; GPA: 4.65/5.00

Current Research

The Unequal Returns to AI Adoption: Productivity and Markups in the Chinese and U.S. Industrial Sectors

- Examines the divergent effects of AI adoption on productivity and markups using listed-firm data from the Chinese and U.S. industrial sectors from 2015 to 2024.
- Combines text-based measures of AI adoption, difference-in-differences estimation, a conceptual framework, and a Kalman-filtered Itô process to project future AI expansion.
- Finds that AI primarily raises markups in China but improves productivity in the United States, reflecting differences in adoption costs and the depth of organizational integration.

The Economic Value of Precision Agriculture under Drought Risk: Evidence from U.S. Corn Counties

- Examines whether precision agriculture mitigates the effects of drought on corn yields and crop insurance losses using county-level data from major U.S. corn-producing states.
- Combines county-year panel models, entropy balancing, staggered difference-in-differences, machine learning, and a dynamic stochastic welfare framework.
- Finds that precision agriculture reduces drought-related production and insurance losses and generates estimated benefits of approximately $5–$20 per acre per year.

Dissertation

Theme: Geopolitical Risk, Technology Catch-up, Energy Transition, and Food Security

- Food Security Concerns under Geopolitical Risk: Land, Trade, and Ecology.
- Geopolitical Risk and Renewable Energy Transition.
- U.S.–China Technological Decoupling and Indigenous Innovation in China.

Peer-Reviewed Publications

- **Dai, S.*** & Ji, J. (2026). Unintended carbon cost of automation technology in transforming economies: The role of capital dependence and structure change. *Ecological Economics.*
- **Dai, S.***, Ji, J., & Yu, H. (2026). Technology Import and Biased Technological Progress in Transforming Economies: Evidence From China. *Review of Development Economics.*
- Yang, B., Fan, F.*, **Dai, S.***, et al. (2026). Industrial Robot, Resource Misallocation, and Firm Innovation Performance. *Journal of Regional Science.*
- **Dai, S.*** (2025). Understanding automation's impact on ecological footprint. *Environmental and Resource Economics.*
- Li, G., Zhang, N., **Dai, S.**, et al. (2025). Trade liberalization, factor network association, and energy poverty. *Energy Economics.*
- Cao, Y. & **Dai, S.*** (2025). Understanding and evaluating new quality productive forces in China: based on machine learning. *Applied Economics Letters.*
- **Dai, S.**, Dai, Y., Hu, J., & Yu, H. (2025). Energy-saving technological change, regional economy growth and endowment structure: empirical evidence from Chinese cities. *Technology Analysis & Strategic Management*, 1–14.
- Fan, F., Yang, B., Cao, Y., & **Dai, S.*** (2024). The spatial effects of innovation agglomeration on air quality in China: from the perspective of industrial development. *International Journal of Sustainable Development & World Ecology*, 1–18.
- Yu, H., **Dai, S.**, & Ke, H. (2024). Industrial collaborative agglomeration and green economic efficiency—Based on the intermediary effect of technical change. *Growth and Change*, 55(2), e12727.
- **Dai, S.**, Yu, H., Aiya, F., & Yang, B. (2024). Green bonds and sustainable development: theoretical model and empirical evidence from Europe. *International Journal of Sustainable Development & World Ecology*, 1–16.
- **Dai, S.**, Dai, Y., & Yu, H. (2024). The effect of gender gap in labor market participation on carbon emission efficiency: state level empirical evidence from the US. *Energy and Environment*, 1–27.
- Fan, F., Zhang, K., **Dai, S.**, & Wang, X. (2023). Decoupling analysis and rebound effect between China's urban innovation capability and resource consumption. *Technology Analysis & Strategic Management*, 35(4), 478–492.
- Fan, F., **Dai, S.**, Yang, B., & Ke, H. (2023). Urban density, directed technological change, and carbon intensity: An empirical study based on Chinese cities. *Technology in Society*, 72, 102151.
- Ke, H., **Dai, S.**, & Fan, F. (2022). Does innovation efficiency inhibit the ecological footprint? An empirical study of China's provincial regions. *Technology Analysis & Strategic Management*, 34(12), 1369–1383.
- Ke, H., Yang, B., & **Dai, S.*** (2022). Does intensive land use contribute to energy efficiency?—evidence based on a spatial durbin model. *International Journal of Environmental Research and Public Health*, 19(9), 5130.
- Ke, H., **Dai, S.**, & Yu, H. (2022). Effect of green innovation efficiency on ecological footprint in 283 Chinese Cities from 2008 to 2018. *Environment, Development and Sustainability*, 24(2), 2841–2860.

- Fan, F., **Dai, S.**, Zhang, K., & Ke, H. (2021). Innovation agglomeration and urban hierarchy: Evidence from Chinese cities. *Applied Economics*, 53(54), 6300–6318.
- Ke, H., **Dai, S.**, & Yu, H. (2021). Spatial effect of innovation efficiency on ecological footprint: City-level empirical evidence from China. *Environmental Technology & Innovation*, 22, 101536.

** Corresponding author.*

Working Papers

- **Dai, S.**, Ji, J., & Huang, K.* Mitigating the Heat: The Role of AI Technology in Shielding Corporate Environmental Performance from Extreme Temperatures. *R&R at Business Strategy and the Environment.*
- Yu, H. & **Dai, S.*** Rural E-commerce and land-augmenting technological progress. *R&R at Land Use Policy.*
- **Dai, S.**, Ji, J., & Huang, K.* Signaling Corporate Resilience under Extreme Weather Shocks: Strategy, Visibility, and Market Responses. *R&R at Business Strategy and the Environment.*
- **Dai, S.**, Lu, C.*, & Li, G.* Pollution-Induced (Mis-)Reallocation in Research: Evidence from U.S. Water Quality Shocks. *R&R at The B.E. Journal of Economic Analysis & Policy.*

Teaching Experience

PhD Math Camp — Instructor, Summer 2025
University of Florida (FRE); Evaluation: 4.9/5
Course materials

AEB 3103 Principles of Food and Resource Economics — Teaching Assistant, Fall 2025
University of Florida; Instructor: James Ji

AEB 4138 Advanced Agribusiness Management — Teaching Assistant, Spring 2025
University of Florida; Instructor: Jacklyn Kropp

ECO 7115 Microeconomic Theory — Teaching Assistant, Fall 2024
PhD Core Course, University of Florida; Instructor: Fatma Gunay

AEB 2014 Economic Issues, Food and You — Teaching Assistant, Fall 2023, Spring 2024
University of Florida; Instructor: Jennifer Clark

Presentations

- **2026**: *Berkeley/Sloan Summer School in EEE*, Berkeley (2026.08); *AAEA Annual Conference*, Kansas (2026.07); *CES NA Conference*, Atlanta (2026.03); *SAEA 2026 Conference*, Louisville (2026.2);
- **2025**: *SEA 2025 Conference*, Tampa (2025.11); *CES CN Conference*, Guangzhou (2025.07); *CAERE Annual Conference*, Shanghai (2025.05); *Seminar on Climate Change Economics*, Online (2025.01)
- **2024**: *Seminar on Environmental Economics*, Online (2024.11)
- **2023**: *China Regional Economics Annual Forum*, Nanjing (2023.07)
- **2022**: *Jiangsu Young Economists Forum*, Nanjing (2022.11); *Future Economist Forum*, Beijing (2022.11)
- **2021**: *China Innovation Economic Forum*, Wuhan (2021.06); *Seminar on Urban and Rural Development*, Nanjing (2021.05)
- **2020**: *China Spatial Economics Annual Conference*, Guangzhou (2020.12)

Awards and Honors

- *Sloan Foundation Travel Scholarship*, ARE, UC Berkeley — 2026
- *AEM/GSS Case Study Competition Second Place*, AAEA — 2026
- *J. R. Greenman Memorial Scholarship*, CALS, University of Florida — 2025

- *Harold B. Clark Award*, FRE, University of Florida 2024
- *Grinter Fellowship*, FRE, University of Florida 2023
- *Amerisolar Scholarship* (Top 3), Hopkins Nanjing Center 2023
- *National Scholarship for Graduate Students* (Top 1), Wuhan University 2022

Academic Services

- Ad hoc reviewer for more than 20 journals, including: *Nature Cities*, *Energy Economics*, *Journal of Business Research*, *Technovation*, *Cities*, *International Review of Financial Analysis*, *Technology in Society*, *Growth and Change*, and *Transportation Research Part E.*
- Academic reputation survey respondent for *Times Higher Education* and *U.S. News & World Report.*

Professional Service, Outreach, and Experience

- *Leadership*: *Academic Vice President, AE-Graduate Student Organization* (2025–2026).
- *Outreach Publication*: Coauthored "Diverging Paths: How Latin America Is Reorienting Exports amid Geopolitical Tensions," *Choices Magazine* (2026).
- *Volunteer*: *UF Water Institute Symposium* (2024), *World Laureates Forum II* (2019).
- *Internship*: *CITIC Construction Investment Co., Ltd.* (2019).

Skills

- *Languages*: English, Chinese, French (Basic).
- *Programming*: R, Python, Stata, MATLAB, SQL, LaTeX.

Professional Memberships

- American Economic Association (AEA); Agricultural & Applied Economics Association (AAEA); Chinese Economist Society (CES)

References

Dr. Zhengfei Guan (Chair)
Associate Professor
Food and Resource Economics Department
University of Florida
E-mail: guanz@ufl.edu

Dr. Fan Fan (Committee Member)
Assistant Professor
Food and Resource Economics Department
University of Florida
E-mail: ffan@ufl.edu

Dr. Gunnar Heins (Committee Member)
Associate Professor
Department of Economics
University of Florida
E-mail: gheins@ufl.edu

Dr. James Ji (Committee Member)
Assistant Professor
Department of Food, Agricultural and Resource Economics
University of Guelph
E-mail: james.ji@uoguelph.ca

Dr. Kurt Stein (External Referee)
Associate Professor of Practice
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Virginia Tech
E-mail: kstein@vt.edu

Dr. Patrick Ward (Committee Member)
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